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A CONTAINER HANDLER, A STORAGE AND RETRIEVAL SYSTEM COMPRISING THE CONTAINER HANDLER AND A METHOD FOR HANDLING A CONTAINER BY MEANS OF THE CONTAINER HANDLER

Abstract

The invention concerns a container handler (**2**) for bidirectional transfer of containers (**20, 106**) between a port (**205**) of a storage grid (**104**) of an automated storage and retrieval system (**1**) and a conveyor (**210**) for transporting containers. The container handler (**2**) comprises an arm (**215**) having an axis of rotation (AR), an arm displacer (**220**), at least a first coupler (**225***a*) provided at a first end of said arm (**215**), said first coupler (**225***a*) for releasable engagement with a container (**20, 106**). The arm displacer (**220**) is configured to rotate the arm (**215**) about the axis of rotation (AR) with at least the first coupler (**225***a*) held in a horizontal plane between a first and a second arm rotational position and displace the arm (**215**) with the first coupler (**225***a*) in a vertical direction between a first and a second vertical position. The invention further concerns an automated storage and retrieval system (**1**) comprising the container handler (**2**) and a method for handling a container (**20, 106**) by means of the container handler (**2**).

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Background/Summary

FIELD OF THE INVENTION

[0001] On a general level the present invention relates to a container handler and a storage and retrieval system comprising the container handler.

BACKGROUND AND PRIOR ART

[0002] FIG. **1** discloses a prior art automated storage and retrieval system **1** with a framework structure **100** and FIGS. **2**, **3***a*-**3***b* disclose three different prior art container handling vehicles **201**, **301**, **401** suitable for operating on such a system **1**.

[0003] The framework structure **100** comprises upright members **102** and a storage volume comprising storage columns **105** arranged in rows between the upright members **102**. In these storage columns **105** storage containers **106**, also known as bins, are stacked one on top of one another to form container stacks **107**. The members **102** may typically be made of metal, e.g. extruded aluminum profiles.

[0004] The framework structure **100** of the automated storage and retrieval system **1** comprises a rail system **108** arranged across the top of framework structure **100**, on which rail system **108** a plurality of container handling vehicles **301**, **401** may be operated to raise storage containers **106** from, and lower storage containers **106** into, the storage columns **105**, and also to transport the storage containers **106** above the storage columns **105**. The rail system **108** comprises a first set of parallel rails **110** arranged to guide movement of the container handling vehicles **301**, **401** in a first direction X across the top of the frame structure **100**, and a second set of parallel rails **111** arranged perpendicular to the first set of rails **110** to guide movement of the container handling vehicles **301**, **401** in a second direction Y which is perpendicular to the first direction X. Containers **106** stored in the columns **105** are accessed by the container handling vehicles **301**, **401** through access openings **112** in the rail system **108**. The container handling vehicles **301**, **401** can move laterally above the storage columns **105**, i.e. in a plane which is parallel to the horizontal X-Y plane.

[0005] The upright members **102** of the framework structure **100** may be used to guide the storage containers during raising of the containers out from and lowering of the containers into the columns **105**. The stacks **107** of containers **106** are typically self-supportive.

[0006] Each prior art container handling vehicle **201**, **301**, **401** comprises a vehicle body **201***a*, **301***a*, **401***a* and first and second sets of wheels **201***b*, **201***c*, **301***b*, **301***c*, **401***b*, **401***c* which enable lateral movement of the container handling vehicles **201**, **301**, **401** in the X direction and in the Y direction, respectively. In FIGS. **2**-**3***b*, two wheels in each set are fully visible. The first set of wheels **201***b*, **301***b*, **401***b* is arranged to engage with two adjacent rails of the first set **110** of rails, and the second set of wheels **201***c*. **301***c*, **401***c* is arranged to engage with two adjacent rails of the second set **111** of rails. At least one of the sets of wheels **201***b*, **201***c*, **301***b*, **301***c*, **401***b*, **401***c* can be lifted and lowered, so that the first set of wheels **201***b*. **301***b*, **401***b* and/or the second set of wheels **201***c*, **301***c*, **401***c* can be engaged with the respective set of rails **110**, **111** at any one time.

[0007] Each prior art container handling vehicle **201**, **301**, **401** also comprises a lifting device **304**, **404** (visible in FIGS. **3***a*-**3***b*) having a lifting frame part **304***a* for vertical transportation of storage containers **106**, e.g. raising a storage container **106** from, and lowering a storage container **106** into, a storage column **105**. Lifting bands **404***a* and guiding pins **405** are also shown in FIG. **3***b*. The lifting device **304**, **404** comprises one or more gripping/engaging devices which are adapted to engage a storage container **106**, and which gripping/engaging devices can be lowered from the vehicle **201**, **301**, **401** so that the position of the gripping/engaging devices with respect to the vehicle **201**, **301**, **401** can be adjusted in a third direction Z (visible for instance in FIG. **1**) which is orthogonal the first direction X and the second direction Y. Parts of the gripping device of the container handling vehicles **301**, **401** are shown in FIGS. **3***a* and **3***b* indicated with reference numbers **304** and **404**. The gripping device of the container handling device **201** is located within the vehicle body **201***a* in FIG. **2**.
[0008] Conventionally, and also for the purpose of this application, Z=1 identifies the uppermost layer available for storage containers below the rails **110**, **111**, i.e. the layer immediately below the rail system **108**, Z=2 the second layer below the rail system **108**, Z=3 the third layer etc. In the exemplary prior art disclosed in FIG. **1**, Z=8 identifies the lowermost, bottom layer of storage containers. Similarly, X=1 . . . n and Y=1 . . . n identifies the position of each storage column **105** in the horizontal plane. Consequently, as an example, and using the Cartesian coordinate system X, Y, Z indicated in FIG. **1**, the storage container identified as **106′** in FIG. **1** can be said to occupy storage position X=18, Y=1, Z=6. The container handling vehicles **201**, **301**, **401** can be said to travel in layer Z=0, and each storage column **105** can be identified by its X and Y coordinates. Thus, the storage containers shown in FIG. **1** extending above the rail system **108** are also said to be arranged in layer Z=0.
[0009] The storage volume of the framework structure **100** has often been referred to as a grid **104**, where the possible storage positions within this grid are referred to as **40**) storage cells within storage columns. Each storage column may be identified by a position in an X- and Y-direction, while each storage cell may be identified by a container number in the X-, Y- and Z-direction.
[0010] Each prior art container handling vehicle **201**, **301**, **401** comprises a storage compartment or space for receiving and stowing a storage container **106** when transporting the storage container **106** across the rail system **108**. The storage space may comprise a cavity arranged internally within the vehicle body **201***a* as shown in FIGS. **2** and **3***b* and as described in e.g. WO2015/193278A1 and WO2019/206487A1, the contents of which are incorporated herein by reference.
[0011] FIG. **3***a* shows an alternative configuration of a container handling vehicle **301** with a cantilever construction. Such a vehicle is described in detail in e.g. NO317366, the contents of which are also incorporated herein by reference.
[0012] The cavity container handling vehicles **201** shown in FIG. **2** may have a footprint that covers an area with dimensions in the X and Y directions which is generally equal to the lateral extent of a storage column **105**, e.g. as is described in WO2015/193278A1, the contents of which are incorporated herein by reference. The term ‘lateral’ used herein may mean ‘horizontal’.
[0013] Alternatively, the cavity container handling vehicles **401** may have a footprint which is larger than the lateral area defined by a storage column **105** as shown in FIG. **3***b* and as disclosed in WO2014/090684A1 or WO2019/206487A1.
[0014] The rail system **108** typically comprises rails with grooves in which the wheels of the vehicles run. Alternatively, the rails may comprise upwardly protruding elements, where the wheels of the vehicles comprise flanges to prevent derailing. These grooves and upwardly protruding elements are collectively known as tracks. Each rail may comprise one track, or each rail may comprise two parallel tracks: in other rail systems **108**, each rail in one direction may comprise one track and each rail in the other perpendicular direction may comprise two tracks. The rail system may also comprise a double track rail in one of the X or Y direction and a single track rail in the other of the X or Y direction. A double track rail may comprise two rail members, each with a

track, which are fastened together.
[0015] WO2018/146304A1, the contents of which are incorporated herein by reference, illustrates a typical configuration of rail system **108** comprising rails and parallel tracks in both X and Y directions.
[0016] In the framework structure **100**, a majority of the columns **105** are storage columns **105**, i.e. columns **105** where storage containers **106** are stored in stacks **107**. However, some columns **105** may have other purposes. In FIG. **1**, columns **119** and **120** are such special-purpose columns used by the container handling vehicles **201**, **301**, **401** to drop off and/or pick up storage containers **106** so that they can be transported to an access station (not shown) where the storage containers **106** can be accessed from outside of the framework structure **100** or transferred out of or into the framework structure **100**. Within the art, such a location is normally referred to as a ‘port’ and the column in which the port is located may be referred to as a ‘port column’ **119**,**120**. The transportation to the access station may be in any direction, that is horizontal, tilted and/or vertical. For example, the storage containers **106** may be placed in a random or a dedicated column **105** within the framework structure **100**, then picked up by any container handling vehicle and transported to a port column **119**, **120** for further transportation to an access station. The transportation from the port to the access station may require movement along various different directions, by means such as delivery vehicles, trolleys or other transportation lines. Note that the term ‘tilted’ means transportation of storage containers **106** having a general transportation orientation somewhere between horizontal and vertical.
[0017] In FIG. **1**, the first port column **119** may for example be a dedicated drop-off port column where the container handling vehicles **201**, **301** can drop off storage containers **106** to be transported to an access or a transfer station, and the second port column **120** may be a dedicated pick-up port column where the container handling vehicles **201**, **301**, **401** can pick up storage containers **106** that have been transported from an access or a transfer station.
[0018] The access station may typically be a picking or a stocking station where product items are removed from or positioned into the storage containers **106**. In a picking or a stocking station, the storage containers **106** are normally not removed from the automated storage and retrieval system **1**, but are, once accessed, returned into the framework structure **100**. A port can also be used for transferring storage containers to another storage facility (e.g. to another framework structure or to another automated storage and retrieval system), to a transport vehicle (e.g. a train or a lorry), or to a production facility.
[0019] A conveyor system comprising conveyors is normally employed to transport the storage containers between the port columns **119**, **120** and the access station.
[0020] If the port columns **119**, **120** and the access station are located at different heights, the conveyor system may comprise a lift device with a vertical component for transporting the storage containers **106** vertically between the port column **119**, **120** and the access station.
[0021] The conveyor system may be arranged to transfer storage containers **106** between different framework structures, e.g. as is described in WO2014/075937A1, the contents of which are incorporated herein by reference.
[0022] When a storage container **106** stored in one of the columns **105** disclosed in FIG. **1** is to be accessed, one of the container handling vehicles **201**, **301**, **401** is instructed to retrieve the target storage container **106** from its position and transport it to the drop-off port column **119**. This operation involves moving the container handling vehicle **201**, **301** to a location above the storage column **105** in which the target storage container **106** is positioned, retrieving the storage container **106** from the storage column **105** using the container handling vehicle's **201**, **301**, **401** lifting device (not shown in FIG. **2** but visible in FIGS. **3***a* and **3***b*), and transporting the storage container **106** to the drop-off port column **119**. If the target storage container **106** is located deep within a stack **107**, i.e. with one or a plurality of other storage containers **106** positioned above the target storage container **106**, the operation also involves temporarily moving the above-positioned storage

containers prior to lifting the target storage container **106** from the storage column **105**. This step, which is sometimes referred to as “digging” within the art, may be performed with the same container handling vehicle that is subsequently used for transporting the target storage container to the drop-off port column **119**, or with one or a plurality of other cooperating container handling vehicles. Alternatively, or in addition, the automated storage and retrieval system **1** may have container handling vehicles **201**, **301**, **401** specifically dedicated to the task of temporarily removing **20**) storage containers **106** from a storage column **105**. Once the target storage container **106** has been removed from the storage column **105**, the temporarily removed storage containers **106** can be repositioned into the original storage column **105**. However, the removed storage containers **106** may alternatively be relocated to other storage columns **105**.

[0023] When a storage container **106** is to be stored in one of the columns **105**, one of the container handling vehicles **201**, **301**, **401** is instructed to pick up the storage container **106** from the pick-up port column **120** and transport it to a location above the storage column **105** where it is to be stored. After storage containers **106** positioned at or above the target position within the stack **107** have been removed, the container handling vehicle **201**, **301**, **401** positions the storage container **106** at **30**) the desired position. The removed storage containers **106** may then be lowered back into the storage column **105** or relocated to other storage columns **105**.

[0024] For monitoring and controlling the automated storage and retrieval system **1**. e.g. monitoring and controlling the location of respective storage containers **106** within the framework structure **100**, the content of each storage container **106** and the movement of the container handling vehicles **201**, **301**, **401** so that a desired storage container **106** can be delivered to the desired location at the desired time without the container handling vehicles **201**, **301**, **401** colliding with each other, the automated storage and retrieval system **1** comprises a control system **500** (shown in FIG. **1**) which typically is computerized and which typically comprises a database for keeping track of the storage containers **106**.

[0025] To facilitate handling of product items stored in the storage container **106**, the product items may be placed in a delivery container for subsequent placement in the storage container **106**. In this way, the delivery container holding the product items may exit the automated storage and retrieval system **1**, while the storage container **106** remains in the system **1**.

[0026] One way of handling storage containers containing product items is disclosed in US2021229917A1. System of US2021229917A1 comprises a storage container transfer vehicle interacting with a storage container lift assembly comprising at least one movable lifting member. The vertically movable lifting member of the assembly is arranged to receive the storage container from said transfer vehicle.

[0027] A further way of handling storage containers containing product items is disclosed in EP3205606A1 featuring a combined area for container handling and storage.

[0028] An objective of the present invention is to provide a system and a method that allows handling of containers, for instance delivery containers placed in storage containers, or storage containers themselves in a time efficient and resource saving manner.

SUMMARY OF THE INVENTION

[0029] The present invention is set forth and characterized in the independent claims, while the dependent claims describe other characteristics of the invention.

[0030] In a first aspect, the invention concerns a container handler for bidirectional transfer of containers between a port of a storage grid of an automated storage and retrieval system, and a conveyor for transporting containers, said container handler comprising: [0031] an arm having an axis of rotation, [0032] an arm displacer, [0033] at least a first coupler provided at a first end of said arm, said first coupler for releasable engagement with a container, [0034] wherein the arm displacer is configured to: [0035] rotate the arm about the axis of rotation with at least the first coupler held in a horizontal plane between a first and a second arm rotational position, and [0036] displace the arm with at least the first coupler in a vertical direction between a first and a second

vertical position.
[0037] By providing a container handler in accordance with first aspect of the invention, a compact container handling solution having small footprint is achieved. In a related context, a constructionally simple container handler is achieved with only a few moving parts. This also results in a robust and reliable solution being considerably less expensive than prior art solutions, such as robotic arms. The constructional simplicity and robustness don't negatively impact the output capacity of the container handler.
[0038] Moreover, the container handler is versatile and may be employed in connection with different port types, such as conveyor ports, swing ports and carousel ports, and different conveyor types, such as roller conveyors and belt conveyors.
[0039] Another aspect of the invention relates to a method for handling a container in accordance with claim **24**. For the sake of brevity, advantages discussed above in connection with the container handler may even be associated with the corresponding method and are not further discussed.
[0040] For the purposes of this application, the term “container handling vehicle” used in “Background and Prior Art”-section of the application and the term “remotely operated vehicle” used in “Detailed Description of the Invention”-section both define a robotic wheeled vehicle operating on a rail system arranged across the top of the framework structure being part of an automated storage and retrieval system.
[0041] The relative terms “upper”, “lower”, “below”, “above”, “higher” etc. shall be understood in their normal sense and as seen in a Cartesian coordinate system. When mentioned in relation to a rail system, “upper” or “above” shall be understood as a position closer to the surface rail system (relative to another component), contrary to the terms “lower” or “below” which shall be understood as a position further away from the rail system (relative another component).

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0042] The following drawings depict exemplary embodiments of the present invention and are appended to facilitate the understanding of the invention. However, the features disclosed in the drawings are for illustrative purposes only and shall not be interpreted in a limiting sense.
[0043] FIG. **1** is a perspective view of a framework structure of a prior art automated storage and retrieval system.
[0044] FIG. **2** is a perspective view of a prior art container handling vehicle having a centrally arranged cavity for carrying storage containers therein.
[0045] FIG. **3***a* is a perspective view of a prior art container handling vehicle having a cantilever for carrying storage containers underneath.
[0046] FIG. **3***b* is a perspective view of a prior art container handling vehicle having an internally arranged cavity for carrying storage containers therein, wherein the cavity is offset from center relative to a principal direction.
[0047] FIG. **4** is a perspective view of a part of an exemplary automated storage and retrieval system according to the invention, comprising a container handler, a carousel-type port and a conveyor for transporting containers.
[0048] FIG. **5** is a perspective view of the container handler of FIG. **4**.
[0049] FIGS. **6**A-**6**F illustrate an operational sequence of the container handler of FIGS. **4** and **5**.
[0050] FIG. **7** is a perspective view of a coupler of a container handler when in engagement with a container.
[0051] FIG. **8** is a perspective view of a container handler in accordance with an alternative embodiment.
DETAILED DESCRIPTION OF THE INVENTION

[0052] In the following, embodiments of the invention will be discussed in more detail with reference to the appended drawings. It should be understood, however, that the drawings are not intended to limit the invention to the subject-matter depicted in the drawings.
[0053] FIG. **4** is a perspective view of a part of an exemplary automated storage and retrieval system according to the invention, comprising a container handler **2**, a carousel-type port **205** and a conveyor **210** for transporting containers.
[0054] The system **1** of FIG. **4** is equal or similar to the prior art system of FIG. **1**, i.e. it comprises a framework structure, a plurality of container handling vehicles moving in X and Y directions on a rail system constituting the topmost part of the framework structure. Each container handling vehicle comprises a lifting device configured to lift/lower a container from/onto a stack within a storage column and to lower/lift the container through any one of port columns **119**, **120** and into the carousel port **205**.
[0055] It is also shown a container handler operator **3**, said container handler **2** for bidirectional transfer of containers between the port **205** of a storage grid of the automated storage and retrieval system **1** and a conveyor **210** for transporting containers, said conveyor **210** also being part of the system **1**. Parts of the container handler **2** will be described in greater detail in connection with FIG. **5**.
[0056] Turning back to FIG. **4**, the container handler **2** is provided at an interface of the port **205** of the storage grid **104** of the automated storage and retrieval system **1** and the conveyor **210** for transporting containers.
[0057] Still with reference to FIG. **4**, the conveyor **210** for transporting containers is suitably positioned with respect to the carousel port **205** such that the arm **215** of the container handler **2** may be aligned with a container tray (not visible in FIG. **4**) of the port **205** of the storage grid **104** of the automated storage and retrieval system **1** and a longitudinal axis of the conveyor **210** for transporting containers.
[0058] In one preferred embodiment, the carousel port **205** rotates substantially concurrently with the displaceable arm **215**. Furthermore, the carousel port **205** and the displaceable arm **215** are arranged to rotate in opposite directions. This facilitates container handoff between the carousel port **205** and the displaceable arm **215** of the container handler **2**.
[0059] For the sake of structural simplicity of the handler **2**, said container tray and the conveyor **210** of the system **1** are typically provided at the same level.
[0060] Still with reference to FIG. **4**, the system **1** may include two types of containers, a delivery container (tote) and a storage container (bin: both shown and discussed in connection with FIG. **7**. The relative sizes and designs of the totes and bins are such that at least one tote may be placed inside the bin. Each tote may contain one or more product items that may be handled by e.g. human and/or robotic operators. In addition to handling totes, the container handler **2** may handle, i.e. transfer, entire bins, holding either totes or loose product items.
[0061] When the container handler **2** and the conveyor **210** form part of the system **1** with a particular configuration of FIG. **4**, an automated process may be achieved that allows retrieval and storing of containers in a highly efficient and resource saving manner. The automated process performed by the container handler **2** may be controlled by a dedicated control system or by a general control system also controlling the operation of the container handling vehicles, the carousel port **205** and the conveyor belt **210**.
[0062] FIG. **5** is a perspective view of the container handler **2** of FIG. **4**. As discussed in connection with FIG. **4**, the container handler **2** is for bidirectional transfer of containers between a port of a storage grid of an automated storage and retrieval system and a transport conveyor. The container handler **2** comprises a stand **250**. The stand has vertical, floor-supported pillars **252*a***, **252*b*** interconnected by a pair of parallel horizontal crossbars **254**, thereby creating a bridge shape under which the conveyor may pass (as shown in FIG. **4**). The stand **250** straddles the port of the storage grid of the automated storage and retrieval system. The pillars **252*a***, **252*b*** may be affixed to the

floor by means of bolts. A control cabinet **253** is attached to one of the pillars **252***a*. Moreover, the container handler **2** comprises a horizontally extending arm **215** connected to the crossbar **254** and having an axis of rotation AR, an arm displacer **220**, a first coupler **225***a* provided at a first end of said arm **215**, said first coupler **225***a* for releasable engagement with a container (not shown in FIG. **5**) and a second coupler **225***b* provided at a second, opposite end of said arm **215**, said second coupler **225***b* also for releasable engagement with a container. The arm displacer **220** is configured to rotate the arm **215** with the first and the second couplers **225***a*, **225***b* about the vertical axis of rotation AR between a first and a second arm rotational position and to displace the arm **215** with the first and the second couplers **225***a*, **225***b* in a vertical direction between a first and a second vertical position.

[0063] By providing such a container handler **2** a compact container handling solution having comparably small footprint is achieved. Still with reference to FIG. **5**, a constructionally simple container handler **2** is achieved with only a few moving parts. This also results in a robust and reliable solution being considerably less expensive than prior art solutions, such as robotic container transfer arms. Here, the constructional simplicity and robustness don't negatively impact the output capacity of the container handler.

[0064] In one embodiment, the arm displacer **220** is configured to rotate the arm **215** in the horizontal plane between the first and the second arm rotational position when the arm with the first and the second couplers **225***a*, **225***b* is in an upper vertical position.

[0065] The arm with the first and the second couplers **225***a*, **225***b* is arranged to reciprocate about the axis of rotation AR in order to avoid entanglement of power and control cables. In a preferred embodiment, an angular stroke of the reciprocal rotation is 180 degrees.

[0066] Still with reference to FIG. **5**, the arm displacer **220** of the container handler **2** comprises a vertical displacement mechanism **230**. The vertical displacement mechanism **230** comprises a vertically guiding part (**232**: shown in FIG. **6**A) and a vertically displaceable support **234** slidably connected with the vertically guiding part. The arm **215** with the first and the second couplers **225***a*, **225***b* is attached to said support **234**. A vertical displacement motor, for example a DC-motor (not visible in FIG. **5**), displaces the support **234** with the attached arm **215** along the vertically guiding part.

[0067] In addition to the vertical displacement mechanism **230**, the arm displacer **220** comprises a rotational displacement mechanism **240**. The rotational displacement mechanism **240** comprises a rotational displacement motor **242** attached to the vertically displaceable support **234**. The motor **242** rotates the arm **215** with the first and the second couplers **225***a*, **225***b* in the horizontal plane between the first and the second arm rotational position. The arm displacer **220**, its parts and operation are more thoroughly discussed in connection with FIGS. **6**A-**6**F.

[0068] In one embodiment, the container handler **2** comprises a movable weight (not shown) for counterbalancing vertical movement of the arm **215** with the first and the second couplers **225***a*, **225***b*.

[0069] FIGS. **6**A-**6**F illustrate a repetitive operational sequence of the container handler **2** shown in FIGS. **4** and **5**. References used in connection with previous FIGS. are used throughout the part of the description corresponding to FIGS. **6**A-**6**F (although not featured in the FIGS. **6**A-**6**F). Throughout the FIGS. **6**A-**6**F, only one of the first and the second couplers **225***a*, **225***b* is in engagement with the container **20**, **106** while said arm **215** is being vertically displaced or rotated. However, it is envisageable to displace/rotate the arm **215** while both couplers engage one container **20**, **106** each.

[0070] In FIG. **6**A, the arm **215** with a first and a second couplers **225***a*, **225***b* has been vertically lowered to a first, lower vertical position. The first coupler **225***a* is being engaged to a delivery container (tote) placed inside a storage container (bin) delivered by a carousel port **205**. At the same time, the second coupler **225***b* is being disengaged from the delivery container. Generally, when the arm **215** is in the first, lower vertical position, engaging process of the first coupler **225***a* occurs simultaneously with the disengaging process of the second coupler **225***b*.

[0071] In FIG. **6**B, the first coupler **225***a* is in engagement with the delivery container **20** whereas the second coupler **225***b* has released the delivery container on the conveyor belt **210**. The arm **215** with the first and the second couplers **225***a*, **225***b* has been vertically hoisted to a second, upper vertical position. Said arm **215** is in a first arm rotational position. Generally, the arm **215** may only rotate between the first and the second rotational position when the arm is in the second, upper vertical position.

[0072] In FIG. **6**C, the arm **215** with first and the second couplers **225***a*, **225***b* is being rotated in a horizontal plane between the first and a second arm rotational position. The first coupler **225***a* is still in engagement with the container.

[0073] In FIG. **6**D, the rotational movement of the arm **215** is completed and the arm **215** is in the second arm rotational position. The first coupler **225***a* is still in engagement with the container. The arm **215**, in particular the second coupler **225***b*, is aligned with a container tray of the carousel port **205**.

[0074] In FIG. **6**E, the arm **215** with the first and the second couplers **225***a*, **225***b* is being vertically lowered towards the first, lower vertical position. The first coupler **225***a* is still in engagement with the container.

[0075] In FIG. **6**F, the arm **215** with the first and the second couplers **225***a*, **225***b* has reached the first, lower vertical position and the container is being disengaged from the first coupler **225***a* while the second coupler **225***b* is being engaged to the container positioned within another storage bin delivered by the carousel port **205**.

[0076] In this context, the container handler **2** is versatile and may be employed in connection with different port types, such as, in addition to carousel ports, conveyor ports and swing ports, as well as different conveyor types, such as roller conveyors of FIG. **6**A-**6**F or belt conveyors.

[0077] FIG. **7** is a perspective view of couplers **225***a*, **225***b* of a container handler, said couplers being attached to the previously discussed, rotatable and vertically displaceable arm **215**. It is also shown a gear **216** mounted to upper face of said arm **215** for rotational motion of the arm **215** as described in connection with FIGS. **6**A-**6**F.

[0078] The proximal coupler **225***a* is shown in engagement with a delivery container **20** partly enclosed by a storage container **106**. Still with reference to the proximal coupler **225***a*, it comprises a coupler arm **258**, arranged perpendicularly to said displaceable arm **215**, a gripper assembly **260** provided at a periphery of the coupler arm **258**, and an actuator system (not shown) operatively connected to the gripper assembly **260** to allow the gripper assembly **260** to releasably engage with the delivery container **20**.

[0079] The actuator system of the coupler comprises a motor, an actuator control system configured to control operation of the motor, and a linkage interconnecting the motor and the gripper assembly. The motor may for example be a DC motor.

[0080] In one embodiment of the actuator system, the linkage of the actuator of each coupler **225***a*, **225***b* comprises a first link connecting the motor and a first gripper paddle ((being obscured by the tote **20** of FIG. **7**)) of a first gripper element **265***a* and a second link connecting the motor and a gripper paddle **270***b* of the second gripper element **265***b*. The motor is configured to displace the first and second links such that horizontally displaceable gripper paddles **270***a*, **270***b* releasably engage with the container **20**.

[0081] The gripper assembly **260** comprises two oppositely provided gripper elements **265***a*, **265***b*, each gripper element **265***a*, **265***b* comprising a horizontally extending gripper rod **266***a*, **266***b* arranged parallel to the arm **215**, each gripper rod **266***a*, **266***b* being fixedly attached to an end of the coupler arm **258**, abutment sensors **268***a*, **268***b*, **268***c*, **268***d* provided at each end of the gripper rod **266***a*, **266***b*, and a horizontally displaceable first and second gripper paddles **270***a*, **270***b* provided adjacent the gripper rod **266***a*, **266***b*, said gripper paddles **270***a*, **270***b* for effecting releasable engagement with the container **20**.

[0082] The purpose of the abutment sensors **268***a*, **268***b*, **268***c*, **268***d* is to register abutment with a rim **22** of the tote **20**. Abutment sensors may e.g. be mechanical sensors such as pressure activated sensors or electronical proximity sensors. In the former case, the extent of the tote abutment sensors should be such that it is ensured that the tote abutment sensors at contact exert pressure on the rim **22**.

[0083] The horizontally displaceable gripper paddles **270***a*, **270***b* each comprise a horizontally extending protrusion **272***a* (its counterpart being obscured by the gripper paddle **270***b* of FIG. **7**) at a lower end, a ledge, a rib or a fold, for exterior insertion into a corresponding recess or aperture **21** of the container **20**.

[0084] The discussed coupler **225***a* is particularly configured to allow releasable engagement to a delivery container (tote) **20** placed in a storage container (bin) **106** arrangement. However, alternative configurations may be envisaged such as couplers adapted to allow releasable engagement with only the bin **106** and/or with both the tote **20** and the surrounding bin **106**.

[0085] FIG. **8** is a perspective view of a container handler **2** in accordance with an alternative embodiment. The embodiment of FIG. **8** shows the container handler **2** for bidirectional transfer of containers. Only significant structural difference of the handler **2** of FIG. **8** when compared with the handler of FIGS. **4-6** is the absence of the second coupler, i.e. only a first coupler **225***a* is present. Further, the arm **215** is adapted to function in a single coupler configuration. For the sake of brevity, functioning and advantages discussed above in connection with the double coupler configuration may even be associated with the handler shown of FIG. **8** and are not further discussed.

[0086] In the preceding description, various aspects of the container handler for handling containers such as delivery containers (totes) and storage containers (bins), an automated storage and retrieval system comprising such a container handler and associated methods have been described with reference to the illustrative embodiments. For purposes of explanation, specific numbers, systems and configurations were set forth in order to provide a thorough understanding of the system and its workings. However, this description is not intended to be construed in a limiting sense. Various modifications and variations of the illustrative embodiment, as well as other embodiments of the system, which are apparent to persons skilled in the art to which the disclosed subject matter pertains, are deemed to lie within the scope of the present invention.

LIST OF REFERENCE NUMBERS

[0087] **1** Automated storage and retrieval system [0088] **2** Container handler [0089] **3** Container handler operator [0090] **20** Delivery container/tote [0091] **21** Tote aperture [0092] **22** Tote rim [0093] **100** Framework structure [0094] **102** Upright members of framework structure [0095] **104** Storage grid [0096] **105** Storage column [0097] **106** Storage container/bin [0098] **106′** Particular position of storage container [0099] **107** Stack [0100] **108** Rail system [0101] **110** Parallel rails in first direction (X) [0102] **112** Access opening [0103] **119** Delivery port column [0104] **120** Receiving port column [0105] **200** Prior art central cavity container handling vehicle [0106] **201** Vehicle body of the container handling vehicle **200** [0107] **202***a* Drive means/wheel arrangement/first set of wheels in first direction (X) [0108] **202***b* Drive means/wheel arrangement/second set of wheels in second direction (Y) [0109] **205** Port [0110] **210** Conveyor [0111] **215** Arm with axis of rotation [0112] **216** Gear [0113] **220** Arm displacer [0114] **225***a* First coupler [0115] **225***b* Second coupler [0116] **230** Vertical displacement mechanism [0117] **232** Vertically guiding part [0118] **234** Vertically displaceable support [0119] **240** Rotational displacement mechanism [0120] **242** Rotational displacement motor [0121] **250** Stand [0122] **252***a* Stand pillar [0123] **252***b* Stand pillar [0124] **253** Control cabinet [0125] **254** Crossbar [0126] **258** Coupler arm [0127] **260** Gripper assembly [0128] **265***a* First gripper element [0129] **265***b* Second gripper element [0130] **266***a* First gripper rod [0131] **266***b* Second gripper rod [0132] **268***a* First abutment sensor [0133] **268***b* Second abutment sensor [0134] **268***c* Third abutment sensor [0135] **268***d* Fourth abutment sensor [0136] **270***a* First gripper paddle [0137] **270***b* Second gripper paddle

[0138] **272***a* First paddle protrusion [0139] **272***b* Second paddle protrusion [0140] **300** Prior art container handling vehicle [0141] **301** Vehicle body of the container handling vehicle **300** [0142] **302***a* Drive means/first set of wheels in first direction (X) [0143] **302***b* Drive means/second set of wheels in second direction (Y) [0144] **400** Prior art container handling vehicle [0145] **401** Vehicle body of the container handling vehicle **401** [0146] **402***a* Drive means/first set of wheels in first direction (X) [0147] **402***b* Drive means/second set of wheels in second direction (Y) [0148] **403** Lifting device [0149] **404** Gripper elements/claws for storage container **106** [0150] **404***a* Lifting bands for storage container **106** [0151] **405** Guiding pins for storage container **106** [0152] X First direction [0153] Y Second direction [0154] Z Third direction [0155] AR Axis of rotation

Claims

1. A container handler for bidirectional transfer of containers between a port of a storage grid of an automated storage and retrieval system and a conveyor for transporting containers, said container handler comprising: an arm having an axis of rotation, an arm displacer, at least a first coupler provided at a first end of said arm, said first coupler for releasable engagement with a container, wherein the arm displacer is configured to: rotate the arm about the axis of rotation with at least the first coupler held in a horizontal plane between a first and a second arm rotational position, and displace the arm with at least the first coupler in a vertical direction between a first and a second vertical position.
2. The container handler in accordance with claim 1, wherein the arm displacer is configured to: rotate the arm with at least the first coupler in the horizontal plane between the first and the second arm rotational position when the arm and at least the first coupler are in an upper vertical position.
3. The container handler in accordance with claim 1, the container handler comprising a second coupler provided at a second, opposite end of said arm, said second coupler for releasable engagement with a container.
4. The container handler in accordance with claim 3, wherein, when said arm is configured to be vertically displaced or rotated, only one of the first and the second couplers is in engagement with the container.
5. The container handler in accordance with claim 3, wherein the arm with the first and the second couplers is arranged to reciprocate about the axis of rotation.
6. The container handler in accordance with claim 5, wherein an angular stroke of the reciprocal rotation is 180 degrees.
7. The container handler in accordance with claim 1, wherein the arm displacer comprises: a vertical displacement mechanism comprising: a vertically guiding part; a vertically displaceable support slidably connected with the vertically guiding part, the arm with at least the first coupler being fixedly attached to said support; and a vertical displacement motor configured to displace the support with the attached arm along the vertically guiding part.
8. The container handler in accordance with claim 7, wherein the arm displacer comprises: a rotational displacement mechanism comprising: a rotational displacement motor attached to the vertically displaceable support and configured to rotate the arm with at least the first coupler in the horizontal plane between the first and the second arm rotational position.
9. The container handler in accordance with claim 1, wherein the container handler is arranged to be provided at an interface of the port of the storage grid of the automated storage and retrieval system and the conveyor for transporting containers.
10. The container handler in accordance with claim 1, wherein the container handler comprises a stand comprising two legs connected by a crossbar, wherein the arm displacer is connected to the crossbar.
11. The container handler in accordance with claim 10, wherein said stand is configured to straddle the port of the storage grid of the automated storage and retrieval system and/or the conveyor for

transporting containers such that the arm of the container handler is aligned with a container tray of the port of the storage grid of the automated storage and retrieval system and/or the conveyor for transporting containers.

12. The container handler in accordance with claim 11, wherein the container tray associated with the port of the automated storage and retrieval system and the conveyor for transporting containers are provided at the same level.

13. The container handler in accordance with claim 1, wherein the port of the automated storage and retrieval system is a carousel port arranged to rotate substantially concurrently with the arm with at least the first coupler.

14. The container handler in accordance with claim 13, wherein the carousel port and the arm with at least the first coupler are arranged to rotate in opposite directions.

15. The container handler in accordance with claim 1, wherein the container handler comprises a movable weight for counterbalancing vertical movement of the arm with at least the first coupler.

16. The container handler in accordance with claim 1, wherein said container has a delivery container stored therein.

17. The container handler in accordance with claim 1, wherein the first coupler comprises a coupler arm, arranged perpendicularly to the arm, a gripper assembly provided at a periphery of the coupler arm, and an actuator system operatively connected to the gripper assembly to allow the gripper assembly to releasably engage with the container.

18. The container handler in accordance with claim 17, wherein the actuator system of the first coupler comprises a rotational displacement motor, an actuator control system configured to control operation of the rotational displacement motor, and a linkage interconnecting the rotational displacement motor and the gripper assembly.

19. The container handler in accordance with claim 18, wherein the gripper assembly comprises two gripper elements arranged at opposite ends of the coupler arm, each gripper element of the two gripper elements comprising: a horizontally extending gripper rod arranged parallel to the arm, said gripper rod being fixedly attached to an end of the coupler arm, an abutment sensor provided at each end of the gripper rod, a horizontally displaceable gripper paddle provided adjacent the gripper rod, said gripper paddle for releasable engagement with the container.

20. The container handler in accordance with claim 19, wherein the linkage of the actuator of the first coupler comprises: a first link connecting the motor and a first gripper paddle of the first gripper element, and a second link connecting the motor and a second gripper paddle of the second gripper element, wherein the motor is configured to displace the first and second links such that the horizontally displaceable gripper paddles releasably engage with the container.

21. The container handler in accordance with claim 19, wherein the horizontally displaceable gripper paddle comprises a horizontally extending protrusion for exterior insertion into a corresponding recess or aperture of the container.

22. An automated storage and retrieval system comprising: a storage grid comprising a framework structure having a plurality of vertical upright members defining a plurality of storage columns for storing stacks of storage containers, wherein a rail system is arranged on the framework structure, the rail system comprising perpendicular rails, the intersections of which form a grid made up of grid cells, the rails defining access openings for a plurality of storage columns, a remotely operated vehicle comprising drive means configured to travel along the rail system and a storage container lifting device for storing and retrieving storage containers through the grid openings, and a container handler for bidirectional transfer of containers between a port of the storage grid of the automated storage and retrieval system and a conveyor for transporting containers.

23. An automated storage and retrieval system in accordance with claim 22, wherein at least one of the storage containers within the framework structure has a delivery container stored therein.

24. A method for handling a container by a container handler comprising an arm having an axis of rotation, an arm displacer, at least a first coupler provided at a first end of said arm, wherein the method comprises: bidirectionally transferring a container between a port of an automated storage and retrieval system and a conveyor for transporting containers by manipulating the container by means of at least the first coupler, the transferring comprising vertically displacing the arm with the first coupler and rotating the arm with at least the first coupler about the rotational axis in a horizontal plane.

25. A method in accordance with claim 24 for handling a container by a container handler comprising an arm having an axis of rotation, an arm displacer, a first coupler provided at a first end of said arm and a second coupler provided at a second, opposite end of said arm, wherein the method comprises: vertically lowering the arm with the first and the second couplers to a first vertical position, engaging a container by means of one of the first and the second couplers, vertically hoisting the arm with first and the second couplers and the engaged container to a second vertical position, rotating the arm with first and the second couplers and the engaged container about the axis of rotation in a horizontal plane between a first and a second arm rotational position, vertically lowering the arm with the first and the second couplers and the engaged container to the first vertical position, and disengaging the container from the one of the first and the second couplers.

26. A method in accordance with claim 25, wherein steps of the method are performed sequentially.

27. A method in accordance with claim 26, wherein the method comprises: disengaging the container from the one of the first and the second couplers while engaging another container with the other one of the first and the second couplers.

28. A method in accordance with claim 25, comprising the step of: counterbalancing vertical movement of the arm with the first and the second couplers by means of a movable weight.

29. A method in accordance with claim 24, wherein the container to be handled is a delivery container (**20**) stored in a storage container (**106**) disposed in the port (**205**) of the automated storage and retrieval system (**1**).

30. A computer-readable medium having stored thereon a computer program for controlling a container handler, the container handler comprising an arm having an axis of rotation, an arm displacer, a first coupler provided at a first end of said arm and a second coupler provided at a second, opposite end of said arm, the computer program comprising instructions to execute: vertically lowering the arm with the first and the second couplers to a first vertical position, engaging a container by means of one of the first and the second couplers, vertically hoisting the arm with first and the second couplers and the engaged container to a second vertical position, rotating the arm with first and the second couplers and the engaged container about the axis of rotation in a horizontal plane between a first and a second arm rotational position, vertically lowering the arm with the first and the second couplers and the engaged container to the first vertical position, and disengaging the container from the one of the first and the second couplers.
